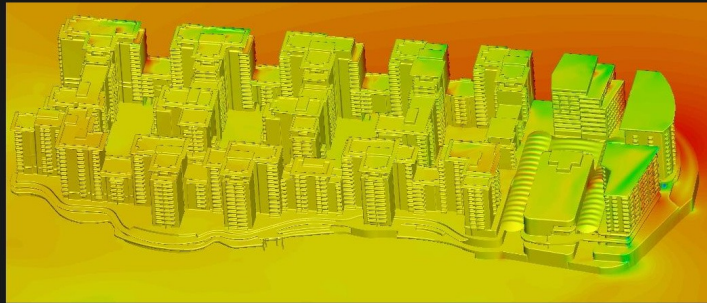


ENGINEERING ANALYSIS

Wind Load and Pedestrian Comfort Assessment

Anonymized Mixed Use Development

Bora Guzel | CFD Analysis and Technical Reporting



CFD overview of the development under the simulated wind field

Original analysis 2022 | Professional English portfolio edition 2026

Executive Summary

This report presents a professional English edition of a 2022 CFD study for an anonymized multi-building development in Istanbul. The original work assessed two connected questions: the wind pressures that the building facades could experience during extreme directional wind events, and the frequency with which wind conditions could become uncomfortable in selected pedestrian, vehicle, and terrace zones.

The wind climate was not reduced to one generic input. REPA wind statistics were separated into eight directions and represented with Rayleigh distributions. Direction-specific 10 m mean speeds and 50-year design speeds were derived using the approach documented in the source study. The applied design speeds ranged from 20.00 m/s from the southeast to 28.10 m/s from the southwest. Terrain Category IV was selected to represent the dense urban surroundings.

The development and nearby buildings were converted into a CFD-ready model and placed in an extended computational domain. Approximately 40 million unstructured tetrahedral cells were used, with local refinement near facades, edges, the ground, and occupied zones. The study used ANSYS software, a k-epsilon turbulence model, and a reported inlet turbulence intensity of 1 percent.

The eight directional pressure cases were enveloped at every facade location. The tabulated facade-zone design pressures ranged from -1.40 kPa suction to +0.90 kPa positive pressure. The comfort assessment divided the project into 17 zones at heights of 1.0, 7.0, 13.5, 19.5, and 25.5 m. Using the study's 6 m/s discomfort threshold and directional wind frequencies, the most exposed ground entrance produced an equivalent 181 uncomfortable days per year. Protected upper terraces produced the lowest values, between 27 and 33 days per year.

Measure	Reported result	Engineering meaning
Directional cases	8	Separate climate and CFD condition for each compass direction
Applied 50-year wind speed	20.00 to 28.10 m/s	Southwest governs the input-speed comparison
Computational mesh	About 40 million cells	Detailed urban model with local refinement
Facade-zone pressure envelope	-1.40 to +0.90 kPa	Minimum suction and maximum positive design values
Comfort zones	17 zones on 5 elevations	Ground routes, vehicle routes, canopies, and terraces
Annual discomfort result	27 to 181 equivalent days	Mitigation priority changes sharply by location

The source study recommended wind mitigation at the JKLMN entrance and the ground-level passages between Blocks A and E. Suitable concepts included porous screens, fences, staggered walls, and dense evergreen planting. Terrace-edge protection was recommended where a higher comfort target was required. These measures should be tested in updated CFD before design adoption because the original model did not document mesh independence, validation, or a current code-compliance review.

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1 Project Definition

1.1 Purpose

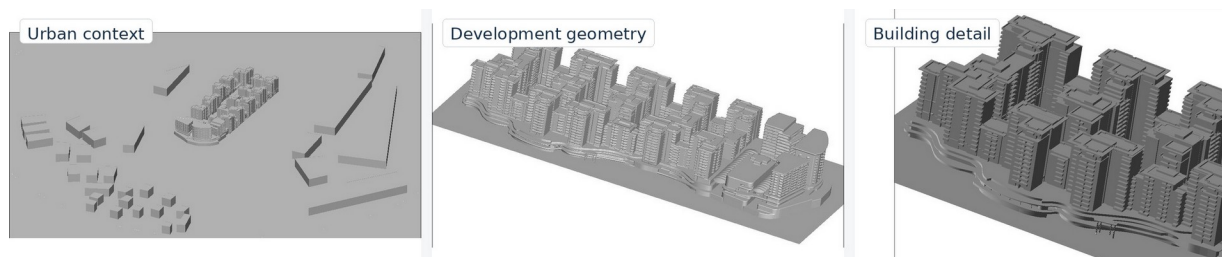
The assessment established design wind pressures for building-envelope development and identified occupied areas where the local wind climate could compromise comfort. Both outputs came from the same urban CFD model but required different treatment. Facade loading used extreme directional wind inputs and an envelope of positive and negative pressures. Comfort used mean directional conditions, local velocity amplification, wind frequency, and an annual probability calculation.

1.2 Study Outputs

1. Derive direction-specific mean and 50-year wind speeds from the available REPA climate record.
2. Build a CFD-ready model of the development and surrounding urban context.
3. Run eight directional cases and envelope facade pressures by building and facade zone.
4. Measure the maximum mean velocity in 17 occupied zones across five elevations.
5. Convert local velocity into an annual discomfort probability using the source study's Rayleigh method and 6 m/s threshold.
6. Identify where windbreaks or additional edge protection would provide the greatest benefit.

1.3 Confidentiality and Use

The client name, development name, and academic approval identity have been removed. Building letters and zone identifiers are retained because they are needed to interpret the results. This edition reconstructs and explains the original analysis; the simulations were not rerun. It is suitable as a portfolio engineering record, not as a current construction document or independent certification of compliance.



Development geometry and surrounding urban context used in the CFD model

2 Wind Climate and Design Inputs

2.1 Design Basis

The source study used TS EN 1991-1-4 as the principal method for the vertical wind profile and 50-year reference event. TS 498, measured Istanbul wind records, the Istanbul High-Rise Buildings Wind Code, and a REPA wind-statistics report were used as comparison sources. Terrain Category IV was selected to represent the dense existing and planned development around the site.

Height	TS 498 reference	Measured comparison	Profile used
8 m	28 m/s	-	30 m/s
10 m	-	25.8 and 27.4 m/s	30 m/s
20 m	36 m/s	-	36 m/s
100 m	42 m/s	-	48 m/s
>100 m	46 m/s	-	Profile continuation

The source treated the 5.9 m/s REPA mean recorded at 30 m as the conservative 10 m reference for the directional statistical calculation. The vertical profile was then generated with the Eurocode formulation for the selected terrain category. This assumption is preserved because no revised meteorological analysis was supplied for this edition.

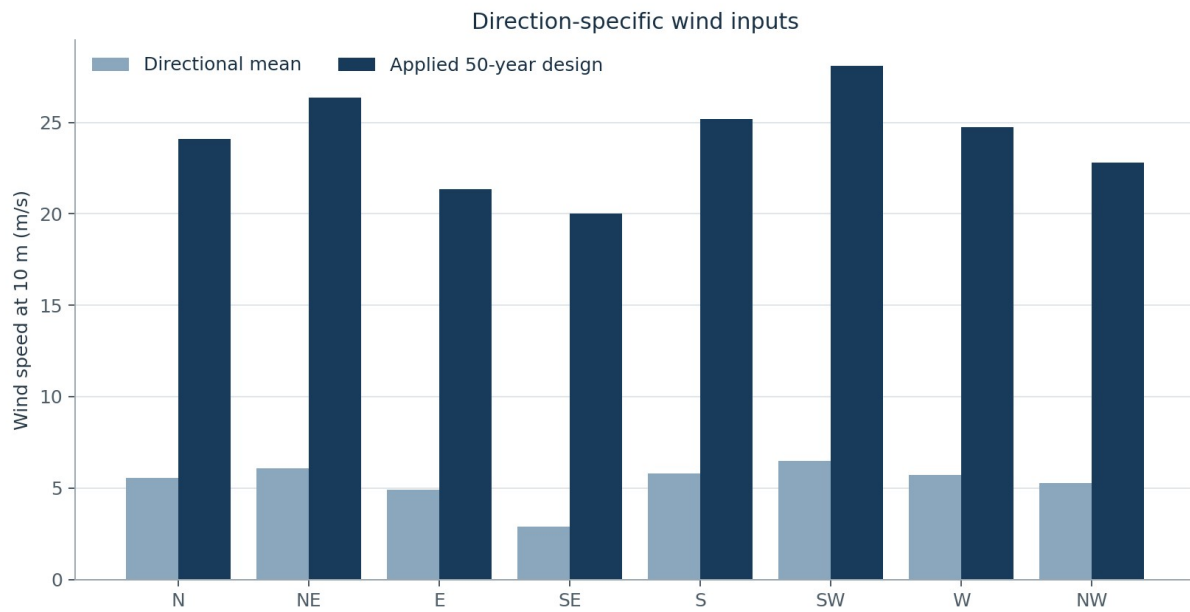
2.2 Directional Statistical Model

A Rayleigh distribution was used to represent wind speed within each direction. In the following expression, v is wind speed and v_m is the directional mean speed.

$$f(v) = \frac{\pi}{2} \frac{v}{v_m^2} \exp \left[-\frac{\pi}{4} \left(\frac{v}{v_m} \right)^2 \right]$$

The source combined the directional frequency and wind-power ratio to recover a mean speed for each direction. The exceedance term from the cumulative distribution was then inverted to estimate the speed corresponding to one 10-minute occurrence in 50 years. The probability used for that event was 1 divided by $50 \times 365 \times 24 \times 6$, or approximately 3.805×10^{-7} .

$$v = v_m \sqrt{\frac{-4 \ln(F)}{\pi}}$$



Directional mean and applied 50-year design wind speeds at 10 m

Direction	Frequency	Mean m/s	Applied design m/s
North	0.103	5.57	24.10
Northeast	0.409	6.08	26.36
East	0.075	4.92	21.36
Southeast	0.034	2.90	20.00*
South	0.054	5.81	25.20
Southwest	0.196	6.48	28.10
West	0.095	5.70	24.73
Northwest	0.035	5.26	22.81

*The computed southeast 50-year speed was 12.57 m/s. The source applied a minimum design input of 20 m/s when the calculated value fell below that threshold.

3 CFD Model and Numerical Method

3.1 Geometry and Domain

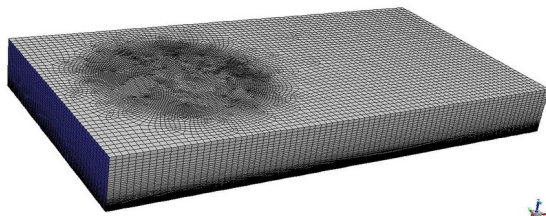
The development was rebuilt as mesh-ready CAD geometry. Nearby buildings were included where they could shield the site, accelerate channelled flow, or alter the wake. The model was placed within an extended domain so the inlet profile could develop before reaching the project and the wake could continue downstream without an immediate outlet constraint.

The source describes a cylindrical transition region around the project and a long wake region. For each wind direction, the inlet and outlet relationship was rotated around the development rather than rebuilding the project geometry. This allowed the same model to support all eight compass cases.

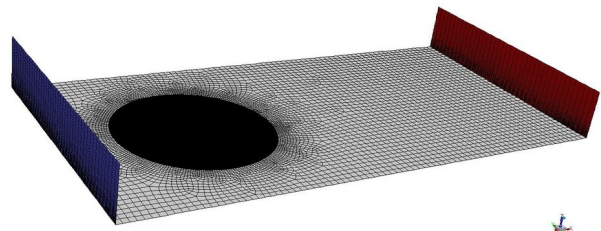
3.2 Mesh Strategy

The model used approximately 40 million unstructured tetrahedral cells. Resolution increased near building faces, facade edges, the ground, and occupied comfort zones where velocity and pressure gradients were expected to be strongest. The outer mesh expanded away from the project so the prescribed boundary profile could be carried into the detailed inner region at a practical computational cost.

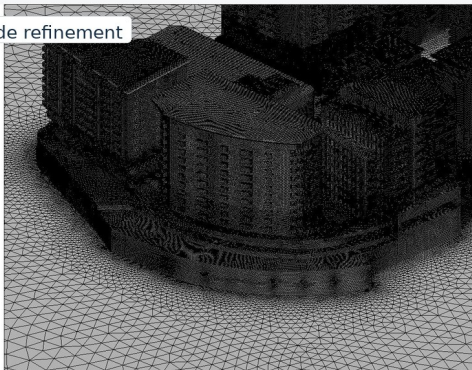
Computational domain



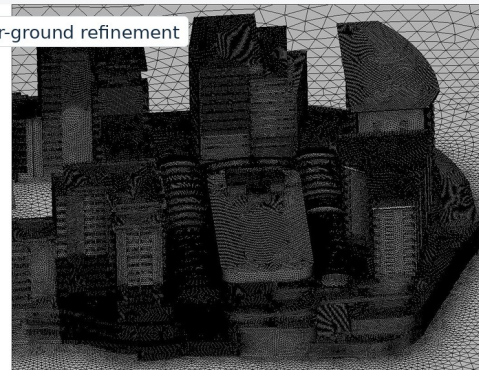
Inlet and outlet extent



Facade refinement



Near-ground refinement



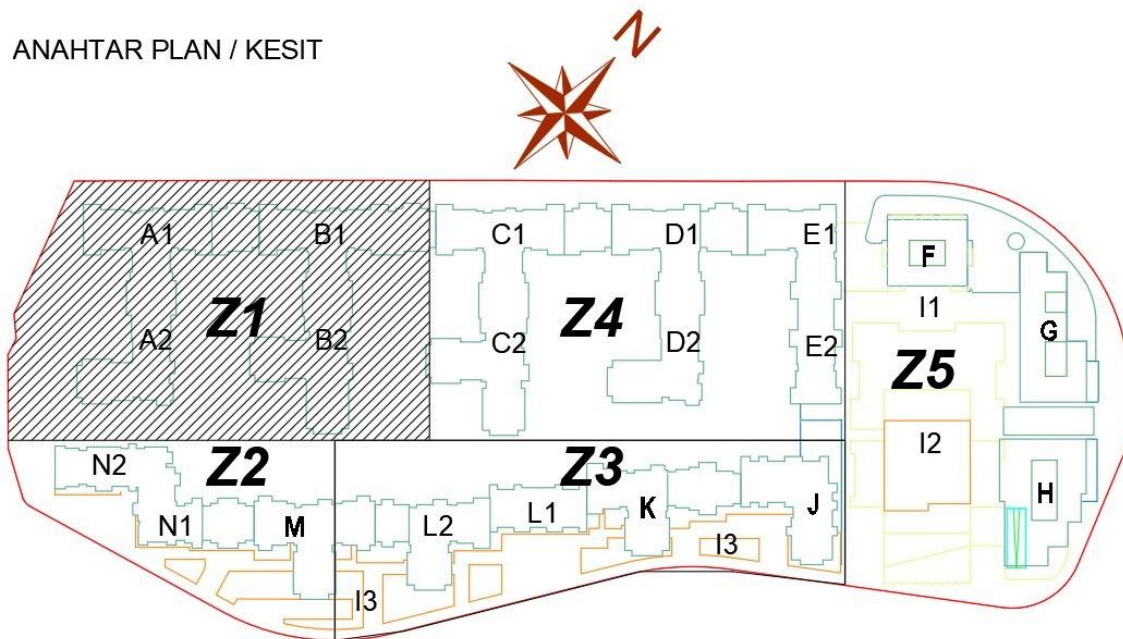
Computational domain and local mesh refinement around the development

3.3 Solver Configuration

Item	Source-study configuration
Numerical method	Finite-volume CFD in ANSYS software
Mesh	Unstructured tetrahedral, approximately 40 million cells
Turbulence model	k-epsilon
Inlet turbulence intensity	1 percent
Wind directions	N, NE, E, SE, S, SW, W, and NW
Computing hardware	32-core HP workstation
Comfort evaluation planes	1.0, 7.0, 13.5, 19.5, and 25.5 m

3.4 Model Output Organization

Building letters and local zone identifiers were retained throughout post-processing so each pressure or comfort result could be traced back to a specific part of the development. This naming system connected the CFD contours, the facade pressure schedule, and the pedestrian comfort assessment without exposing the original project identity.



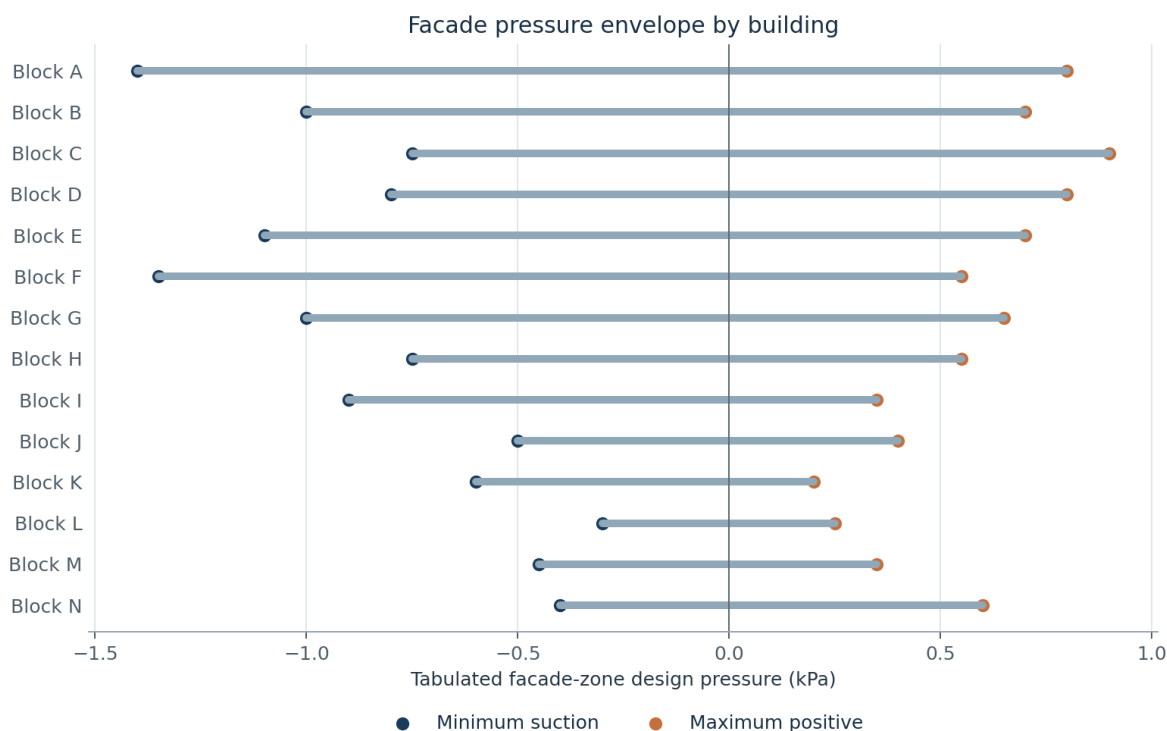
Building and zone identifiers retained from the source model

4 Facade Wind Load Assessment

4.1 Eight-Direction Pressure Envelope

Each building surface was evaluated under the maximum wind condition assigned to all eight directions. The design value at a surface location was not taken from one preferred wind direction. The positive design pressure was the highest pressure obtained across the eight cases, and the negative design pressure was the most severe suction obtained across the same cases.

The source contour scale extends approximately from -2.50 to +1.00 kPa and includes highly local values. The tabulated facade-zone design schedule, which is the appropriate basis for the summarized envelope in this edition, ranges from -1.40 to +0.90 kPa. The difference is retained explicitly rather than combining local contour extrema with the scheduled zone values.

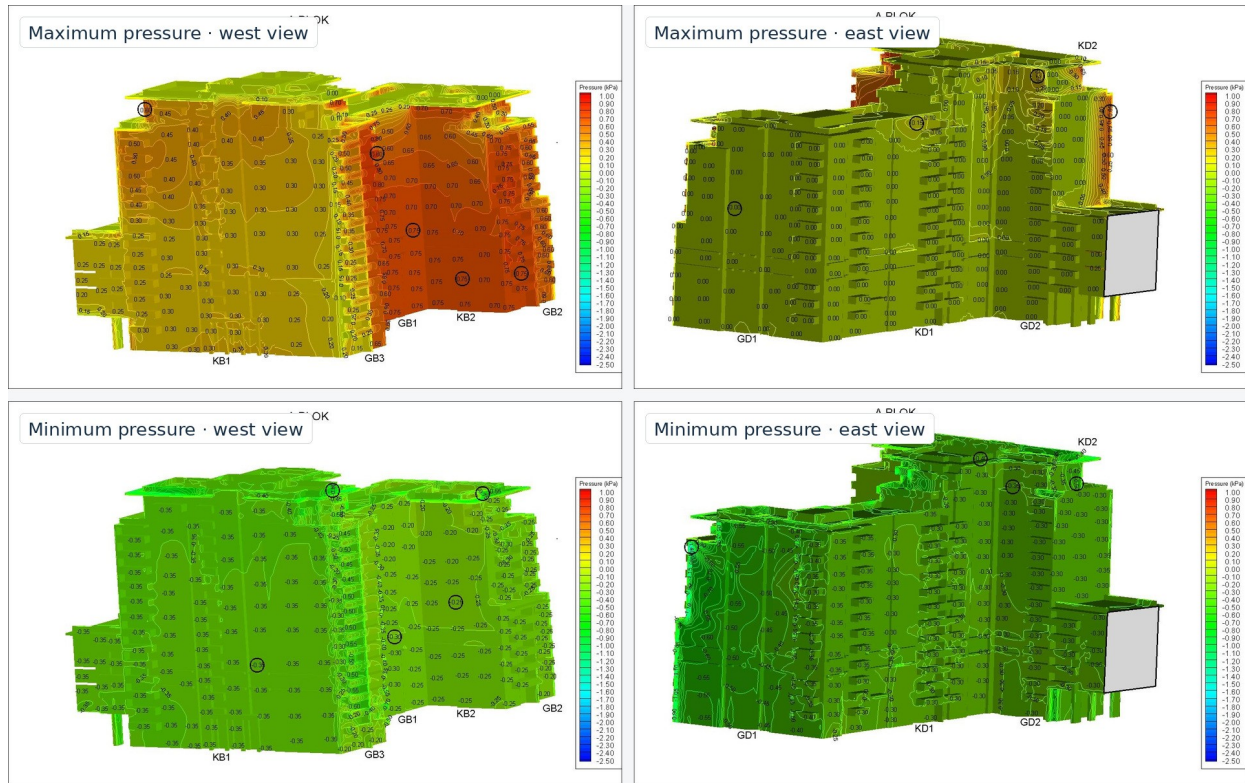


Minimum suction and maximum positive design pressure by building

Result	Value	Location
Governing minimum	-1.40 kPa	Block A, zone GD1
Second-lowest minimum	-1.35 kPa	Block F, zones GB and KB
Governing maximum	+0.90 kPa	Block C, zone GB4
Next-highest maximum	+0.80 kPa	Blocks A and D

4.2 Selected Pressure Contours

Block A illustrates why both signs of pressure are required. Windward surfaces carry positive pressure, while roof edges, returns, recesses, and leeward faces develop suction. The controlling direction changes by facade zone, so a single windward-leeward sketch would not reproduce the design envelope.



Selected maximum and minimum pressure contours for Block A

5 Pedestrian Wind Comfort Assessment

5.1 Comfort Criterion

The source study classified mean wind speed from 0 to 3 m/s as comfortable, 3 to 6 m/s as less comfortable, and values above 6 m/s as uncomfortable. This is a project-specific screening criterion. It is not presented as a substitute for a current Lawson, Davenport, or locally adopted pedestrian-wind standard, which would normally distinguish activities such as sitting, standing, strolling, and fast walking and may use gust-sensitive criteria.

Mean wind speed	Source-study classification
0 to 3 m/s	Comfortable
3 to 6 m/s	Less comfortable
>6 m/s	Uncomfortable

5.2 Annual Probability Method

For each occupied zone and wind direction, the maximum local mean velocity was read from the CFD result. That velocity was treated as v_m in the Rayleigh exceedance equation. The probability of exceeding 6 m/s under that directional condition was then calculated as follows.

$$G(v) = \exp \left[-\frac{\pi}{4} \left(\frac{v}{v_m} \right)^2 \right]$$

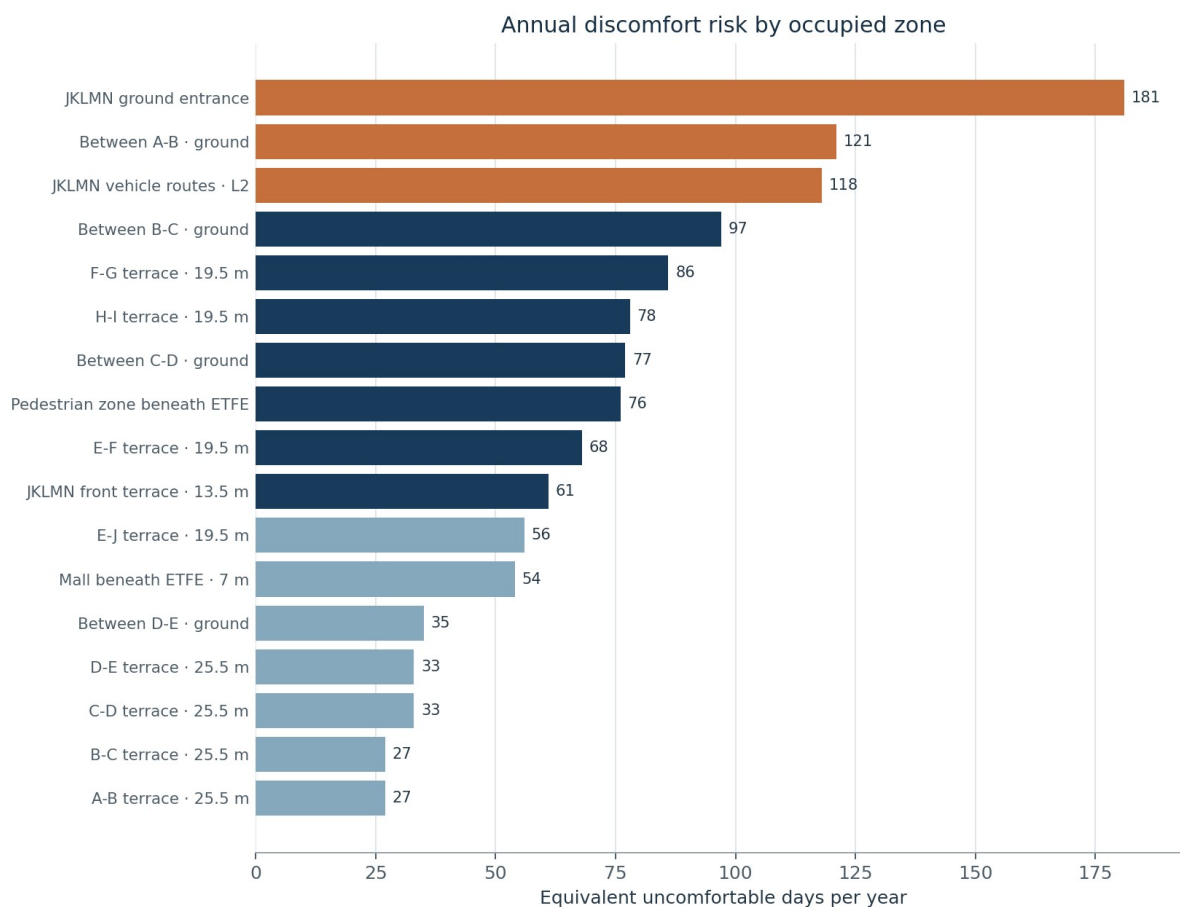
The conditional exceedance probability was multiplied by the frequency of the corresponding wind direction. The eight directional contributions were added and multiplied by 365 to produce an equivalent number of uncomfortable days per year. The result is therefore a statistical exposure indicator, not a list of specific calendar days.



Seventeen comfort zones evaluated on five horizontal planes

5.3 Comfort Results

Exposure was concentrated in three connected circulation areas. The JKLMN ground entrance governed at 181 equivalent uncomfortable days per year. The ground-level route between Blocks A and B produced 121 days, and the Level 2 vehicle routes around JKLMN produced 118 days. At the opposite end of the ranking, the four protected terraces at 25.5 m produced only 27 to 33 days per year under the same method.

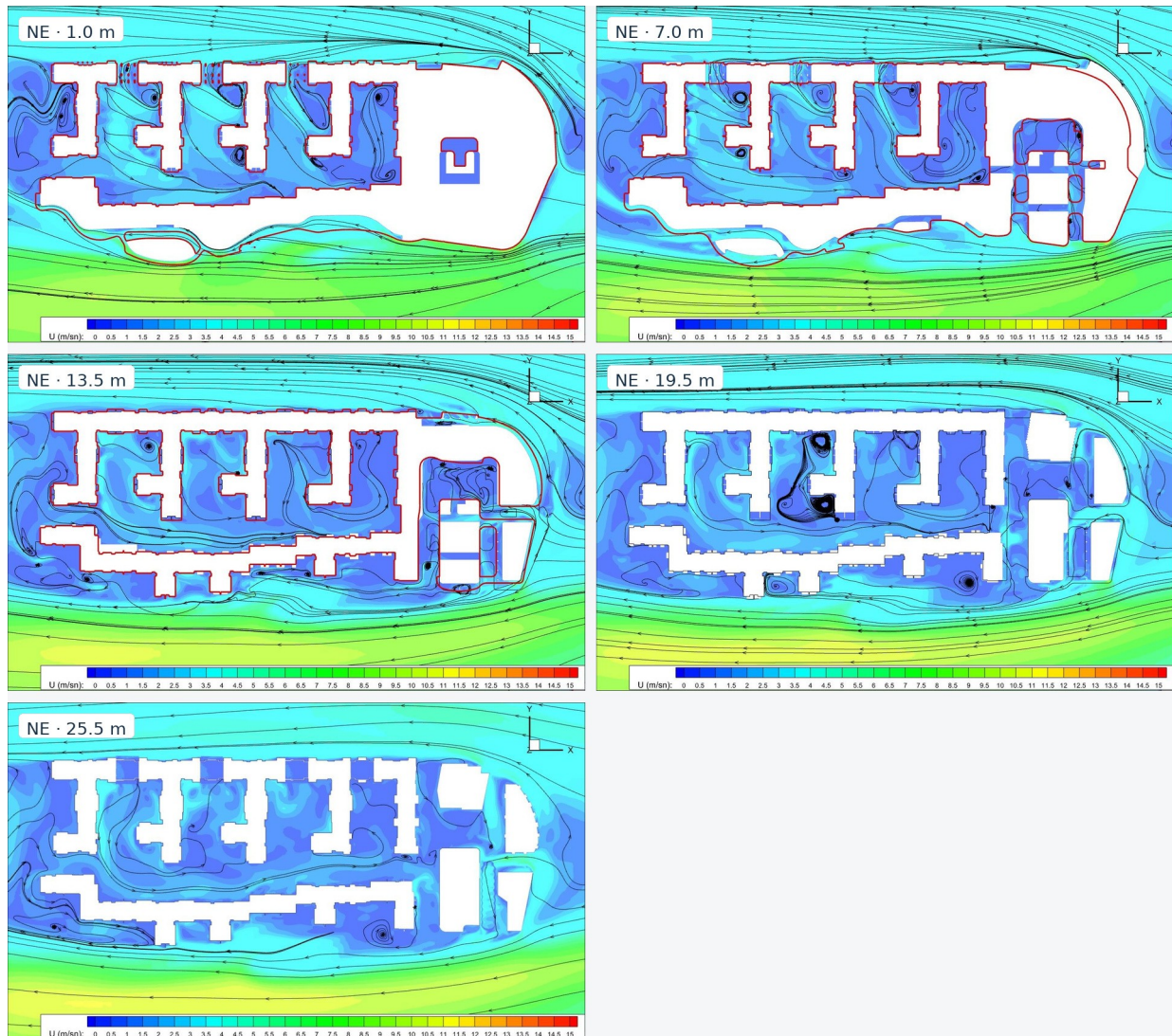


Equivalent uncomfortable days per year calculated for all 17 zones

Priority	Zone	Equivalent days/year	Peak local mean speed
1	JKLMN ground entrance	181	10 m/s from SW
2	Between Blocks A-B at ground level	121	11 m/s from SW
3	JKLMN Level 2 vehicle routes	118	11 m/s from SW
4	Between Blocks B-C at ground level	97	7 m/s from SW
5	Terrace between Blocks F-G	86	9 m/s from N

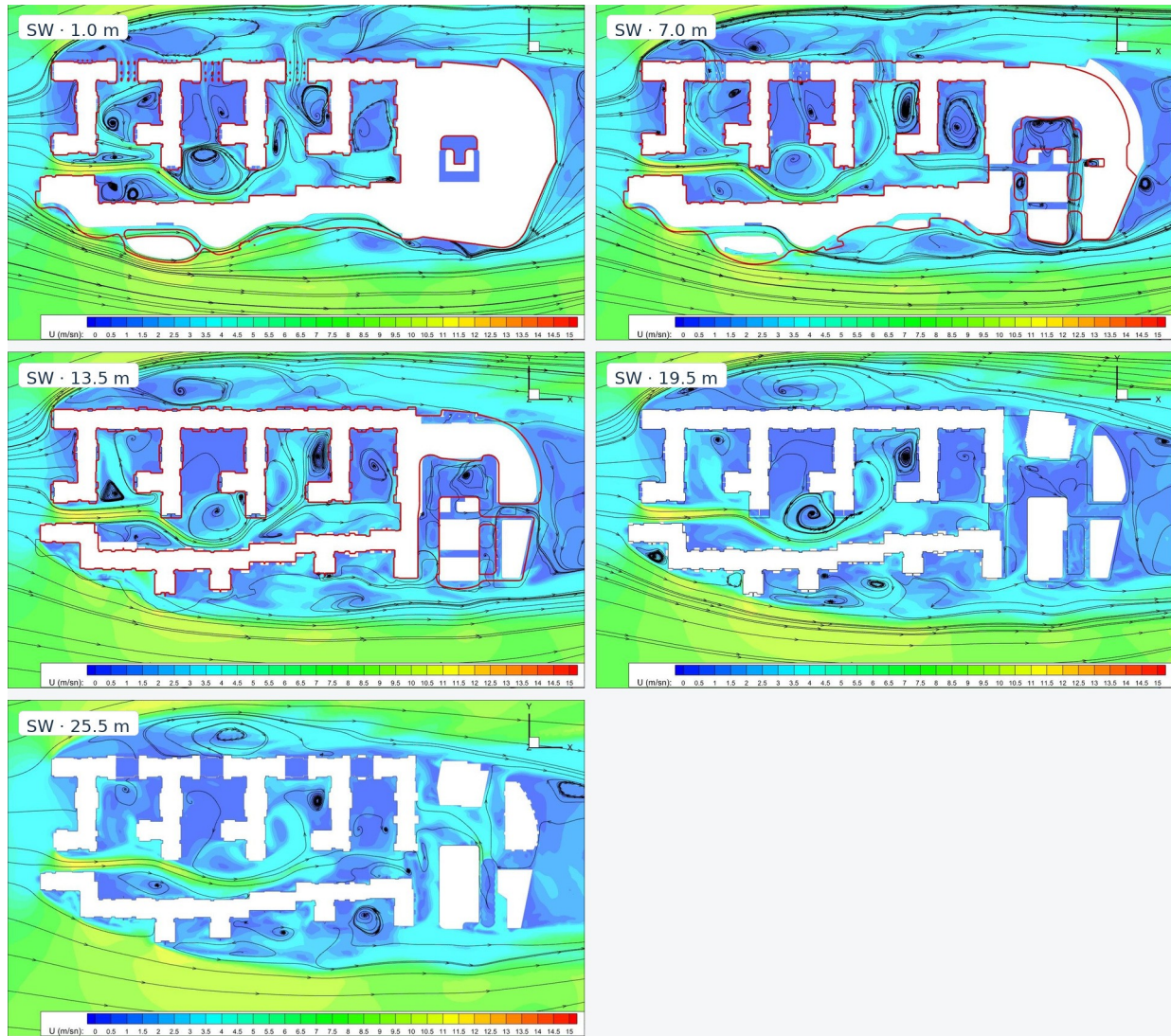
5.4 Directional Flow Fields

Northeast winds were the most frequent in the source climate record, with a directional frequency of approximately 0.409. The plots show channelled flow around the repeated building courts and stronger external flow along the long site edges. As the evaluation plane rises, the lower development geometry shields less of the flow.



Northeast mean-wind velocity fields across the five comfort elevations

Southwest winds combined the highest directional mean speed, the highest applied 50-year design speed, and a substantial frequency of approximately 0.196. These cases generated the peak comfort-zone velocities at the JKLMN entrance, the passage between Blocks A and B, and the Level 2 vehicle routes. The result explains why those areas rise to the top of the annual discomfort ranking even though northeast winds occur more often.



Southwest mean-wind velocity fields across the five comfort elevations

6 Engineering Diagnosis and Recommendations

6.1 Governing Wind Mechanisms

Observation	Evidence	Design implication
Ground entrance is highly exposed	181 equivalent days/year	Treat the JKLMN entrance as the first mitigation priority
Repeated blocks create wind corridors	A-B, B-C, and C-D passages rank high	Interrupt the channel without blocking access or visibility
Southwest winds control several peaks	10 to 11 m/s local means in the highest-risk zones	Test mitigation under SW before broadening to all directions
Some upper terraces are naturally protected	27 to 33 days/year at 25.5 m	Preserve the shielding geometry and existing parapet effect
Facade pressure varies by sign and zone	-1.40 to +0.90 kPa schedule	Carry both suction and positive pressure into envelope development

6.2 Mitigation Recommendations

1. Introduce windbreaks at the JKLMN ground entrance where pedestrian and vehicle movement intersects the most exposed zone.
2. Break up the ground-level wind corridors between Blocks A and E with porous screens, planted barriers, staggered walls, or a coordinated combination of these elements.
3. Avoid selecting a fully solid screen by default. Porous or staggered systems can reduce local acceleration while limiting abrupt separation and preserving visibility.
4. Where a higher comfort target is required on the E-J terrace group, add perimeter wind protection and test its effect on adjacent zones.
5. Retain the shielding function of the approximately 1 m terrace walls reported between Blocks A and E. Consider additional transparent edge protection where safety or a stricter comfort objective requires it.
6. Rerun the eight-direction comfort cases after mitigation. A local fix should not transfer acceleration to the next entrance, terrace, or facade corner.

6.3 Facade Design Use

The pressure schedule should be coordinated with the facade engineer by zone, including corners, returns, roof edges, recessed areas, and transitions between building forms. The values in this report are an explanatory reconstruction of the source study. Final cladding, glazing, anchors, and secondary steel require current load combinations, component-specific tributary areas, internal pressure assumptions, and confirmation of the controlling code edition.

7 Decision Path and Study Limitations

7.1 Decision Path

1. Represent the real context. Include nearby buildings because shielding and channelled flow can control both facade pressure and pedestrian comfort.
2. Separate the climate by direction. Use wind frequency and power information to derive eight mean speeds instead of applying one site-wide value.
3. Create extreme inputs. Convert each directional climate into a 50-year design speed and apply the documented 20 m/s minimum where required.
4. Resolve the geometry where gradients matter. Concentrate mesh near facades, edges, the ground, and occupied zones within an extended domain.
5. Envelope the load cases. Select the highest positive pressure and most severe suction at every facade zone across all eight directions.
6. Convert flow into exposure. Use each zone's local mean velocity and directional frequency to estimate annual discomfort risk.
7. Prioritize mitigation by consequence. Address the entrance and circulation corridors first, then verify that screens or planting do not move the problem elsewhere.

7.2 Limitations Requiring Resolution

- The simulations were not rerun for this edition, and the source report does not provide residual histories, mass-balance checks, or formal convergence criteria.
- A mesh-sensitivity study is not documented. Approximately 40 million cells indicates substantial resolution but does not by itself demonstrate mesh independence.
- The model is not documented as validated against wind-tunnel testing or site measurements.
- The comfort method uses mean velocity and a single 6 m/s threshold. It does not explicitly evaluate gusts, turbulence exposure, or activity-specific comfort categories.
- The annual values are probability-derived equivalent days, not observed days or a time-resolved pedestrian exposure model.
- The reported standards and climate records require confirmation against current requirements before design approval or construction use.
- The source reports both broad contour limits near -2.50 and +1.00 kPa and a facade-zone design schedule of -1.40 to +0.90 kPa. The scheduled values are used for the summary, and the local contour range remains a review item.

7.3 Conclusion

The study linked climate, urban form, facade loading, and human exposure within one model. Its most useful result is not the colour of an individual contour. It is the ranking of decisions: direction-specific wind inputs, a pressure envelope for the facade, and a quantified comfort priority that directs mitigation to the locations where it can matter most. The next engineering step is an updated, validated CFD study of the selected windbreak concepts and the final facade geometry.

References

1. European Committee for Standardization. EN 1991-1-4. Eurocode 1: Actions on structures. Part 1-4: General actions - Wind actions.
2. Kandilli Observatory and Earthquake Research Institute, Bogazici University. Istanbul High-Rise Buildings Wind Code. Version IV, August 2008.
3. Turkish Standards Institution. TS 498. Design loads to be considered in the dimensioning of structural elements.
4. Envirometrics Inc. Meeting the IBC Wind Loading Study Requirements. Reference cited by the source study.
5. REPA wind-statistics record used by the source study for a representative Istanbul location. Exact coordinate withheld with the anonymized project record.

Appendix A Facade Pressure Summary

The table below summarizes the governing tabulated facade-zone pressures for each building. Zone codes are retained from the source model so they continue to correspond with the contour labels. They are identifiers, not a replacement for a current facade elevation and load map.

Building	Min zone	Min kPa	Max zone	Max kPa
Block A	GD1	-1.40	GB3	0.80
Block B	KB1	-1.00	GB2	0.70
Block C	KD2	-0.75	GB4	0.90
Block D	KD	-0.80	GB3	0.80
Block E	KB	-1.10	GB2	0.70
Block F	GB	-1.35	KB	0.55
Block G	KD	-1.00	KD	0.65
Block H	KD	-0.75	KD	0.55
Block I	KB	-0.90	KD	0.35
Block J	KD	-0.50	KB	0.40
Block K	KB	-0.60	KD2	0.20
Block L	GB1	-0.30	GD1	0.25
Block M	GB1	-0.45	KD1	0.35
Block N	GB2	-0.40	KB2	0.60

Appendix B Comfort Zone Results

Peak local mean speed is the highest of the eight directional CFD values for each zone. Equivalent uncomfortable days per year are calculated with the full directional probability method, so the direction of the peak is not necessarily the only or largest contributor to annual exposure.

ID	Comfort zone	Height m	Peak mean m/s	Days/year
1	JKLMN ground entrance	1	10 SW	181
2	Between Blocks A-B at ground level	1	11 SW	121
3	Between Blocks B-C at ground level	1	7 SW	97
4	Between Blocks C-D at ground level	1	6.5 N	77
5	Between Blocks D-E at ground level	1	4 W	35
6	JKLMN Level 2 vehicle routes	7	11 SW	118
7	Mall zone beneath the ETFE canopy	7	6.5 S	54
8	JKLMN front terrace	13.5	6 S	61
9	Pedestrian zone beneath the ETFE canopy	13.5	10 N	76
10	Terrace between Blocks H-I	19.5	10 N	78
11	Terrace between Blocks F-G	19.5	9 N	86
12	Terrace between Blocks E-F	19.5	9 N	68
13	Terrace between Blocks E-J	19.5	4 W	56
14	Terrace between Blocks A-B	25.5	4.5 N	27
15	Terrace between Blocks B-C	25.5	4.5 N	27
16	Terrace between Blocks C-D	25.5	4.5 N	33
17	Terrace between Blocks D-E	25.5	4.5 N	33